SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Scenario-Based Questions (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 2: Scenario-Based Questions
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	B//2024	**Date:**	28/10/26

Code of Conduct

I _____ (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Scenario-Based Qns

Q1. Image Enhancement Using Grey-Level Transformations

A hospital uses a digital imaging system to capture X-ray images, but some images appear too dark to identify important anatomical structures. The image processing team decides to apply grey-level transformation techniques to improve image visibility. Apply a suitable transformation to enhance the brightness and contrast of the image. Calculate the transformed intensity values for selected pixels using the chosen transformation. Explain how the transformation improves the visibility of important features in the X-ray image.

Q2. Histogram Processing and Spatial Filtering

A security surveillance system captures images at night, resulting in poor contrast and unwanted noise. The system must enhance the images to identify people and objects more clearly. Apply histogram equalization to improve the contrast of the captured image. Use an appropriate spatial filtering technique to reduce noise while preserving important details. Compare the effects of both techniques and explain how they improve the quality of the surveillance images.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

A satellite imaging system captures land images containing noise and blurred boundaries between different geographical regions. The image processing team decides to use frequency domain filtering to improve image quality. Apply a low-pass filter to reduce high-frequency noise and smooth the image. Use a high-pass filter to enhance edges and sharpen important geographical features. Explain the differences between the two filters and determine how each technique contributes to image interpretation.

Q4. Image Segmentation Using Edge Detection and Thresholding

An automated manufacturing system must inspect metal components to identify cracks and surface defects. The captured images contain variations in brightness, making defect detection difficult. Apply an appropriate edge detection technique to identify the boundaries of possible cracks. Use thresholding to separate defective regions from the background and construct a binary image representing the detected defects. Explain how the selected techniques help the inspection system identify defective components.

Q5. Region-Based Segmentation and Boundary Detection

An agricultural monitoring system uses aerial images to identify crop fields, water bodies, and surrounding land regions. The captured images contain regions with similar intensity values and some discontinuous boundaries. Apply region-based segmentation to group neighbouring pixels with similar characteristics. Use edge linking and boundary detection techniques to connect broken edges and identify the boundaries of different regions. Explain how these techniques help distinguish agricultural fields and improve the accuracy of land-region identification.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Image Enhancement Using Grey-Level Transformations

Aim: To improve the brightness and contrast of a dark X-ray image using a grey-level transformation.

Technique: Log transformation.

Formula: $s = c \log(1 + r)$, where r is the input intensity, s is the output intensity, and $c = 255 / \log(256)$.

Calculation (approximate values):

Input intensity (r)	Output intensity (s)
0	0
20	133
50	179
100	212
150	232
255	255

Procedure:

Read the dark X-ray image.

Apply the log transformation to each pixel.

Normalize the output intensity values to the range 0–255.

Display the enhanced image.

Result: Log transformation increases the brightness of darker pixels and makes hidden anatomical structures more visible, helping doctors examine the X-ray image.

Q2. Histogram Processing and Spatial Filtering

Aim: To improve night surveillance images using histogram equalization and spatial filtering.

Techniques: Histogram equalization improves contrast; median filtering reduces noise while preserving edges.

Formula: $s_k = \text{round}(255 \times \text{CDF}(r_k))$, where CDF is the cumulative distribution function of the image histogram.

Procedure:

Read the low-contrast surveillance image.

Calculate the image histogram.

Apply histogram equalization using the CDF formula.

Apply a 3 × 3 median filter to remove salt-and-pepper noise.

Display and compare the original and processed images.

Comparison:

Histogram equalization	Median filtering
Improves contrast and reveals hidden details.	Removes isolated noise and preserves edges.

Result: Histogram equalization improves the visibility of people and objects, while median filtering reduces unwanted noise and maintains important image details.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

Aim: To reduce noise and enhance geographical boundaries using frequency domain filters.

Techniques: Low-pass filter (LPF) and high-pass filter (HPF).

Fourier transform: $F(u,v) = \mathcal{F}\{f(x,y)\}$

Filtering: $G(u,v) = H(u,v)F(u,v)$

Inverse transform: $g(x,y) = \mathcal{F}^{-1}\{G(u,v)\}$

Procedure:

Read the satellite image.

Apply the Fourier transform.

Apply a low-pass filter to suppress high-frequency components.

Apply a high-pass filter to emphasize high-frequency components.

Use the inverse Fourier transform to obtain the processed images.

Comparison:

Low-pass filter	High-pass filter
Smooths the image and reduces high-frequency noise; may blur fine details.	Enhances edges and boundaries; may amplify noise.

Result: The low-pass filter reduces noise and produces a smoother image, while the high-pass filter highlights boundaries between geographical regions, improving satellite image interpretation.

Q4. Image Segmentation Using Edge Detection and Thresholding

Aim: To detect cracks and surface defects in metal components using edge detection and thresholding.

Techniques: Canny edge detection and binary thresholding.

Thresholding formula: $g(x,y) = 255$ if $f(x,y) \geq T$; otherwise $g(x,y) = 0$, where T is the selected threshold.

Procedure:

Read the metal surface image.

Convert it to grayscale.

Apply Gaussian smoothing to reduce noise.

Apply Canny edge detection to identify possible crack boundaries.

Apply thresholding to separate candidate defects from the background.

Display the resulting binary image.

Result: Edge detection identifies crack boundaries, while thresholding separates candidate defective regions from the background. The combined techniques help automate the inspection of metal components.

Q5. Region-Based Segmentation and Boundary Detection

Aim: To identify agricultural fields, water bodies, and surrounding land using region-based segmentation and boundary detection.

Techniques: Region growing, edge linking, and boundary detection.

Region growing condition: $|I(x,y) - \mu R| < T$, where $I(x,y)$ is the neighbouring pixel intensity, μR is the mean intensity of the region, and T is the similarity threshold.

Procedure:

Read the aerial image.

Select seed pixels from representative regions.

Group neighbouring pixels with similar intensity or colour values using region growing.

Detect edges using an edge detector.

Apply edge linking to connect discontinuous edges.

Extract region boundaries and display the segmented image.

Result: Region growing groups similar pixels into meaningful land regions, while edge linking connects broken boundaries. These techniques improve the identification of crop fields, water bodies, and surrounding land.